

Cybersecurity Lab Report

Step 1: Introduction

The goal of this lab is to perform network scanning and traffic analysis to identify vulnerabilities in a target system.

Tools used:

- Zenmap (Nmap GUI) for host discovery and port scanning.
- Wireshark for packet capture and traffic analysis.

Step 2: Environment Setup

Operating System: Kali Linux (Virtual Machine)

Target IP: 10.0.2.15 (eth0 interface)

Tools Installed: Zenmap, Wireshark

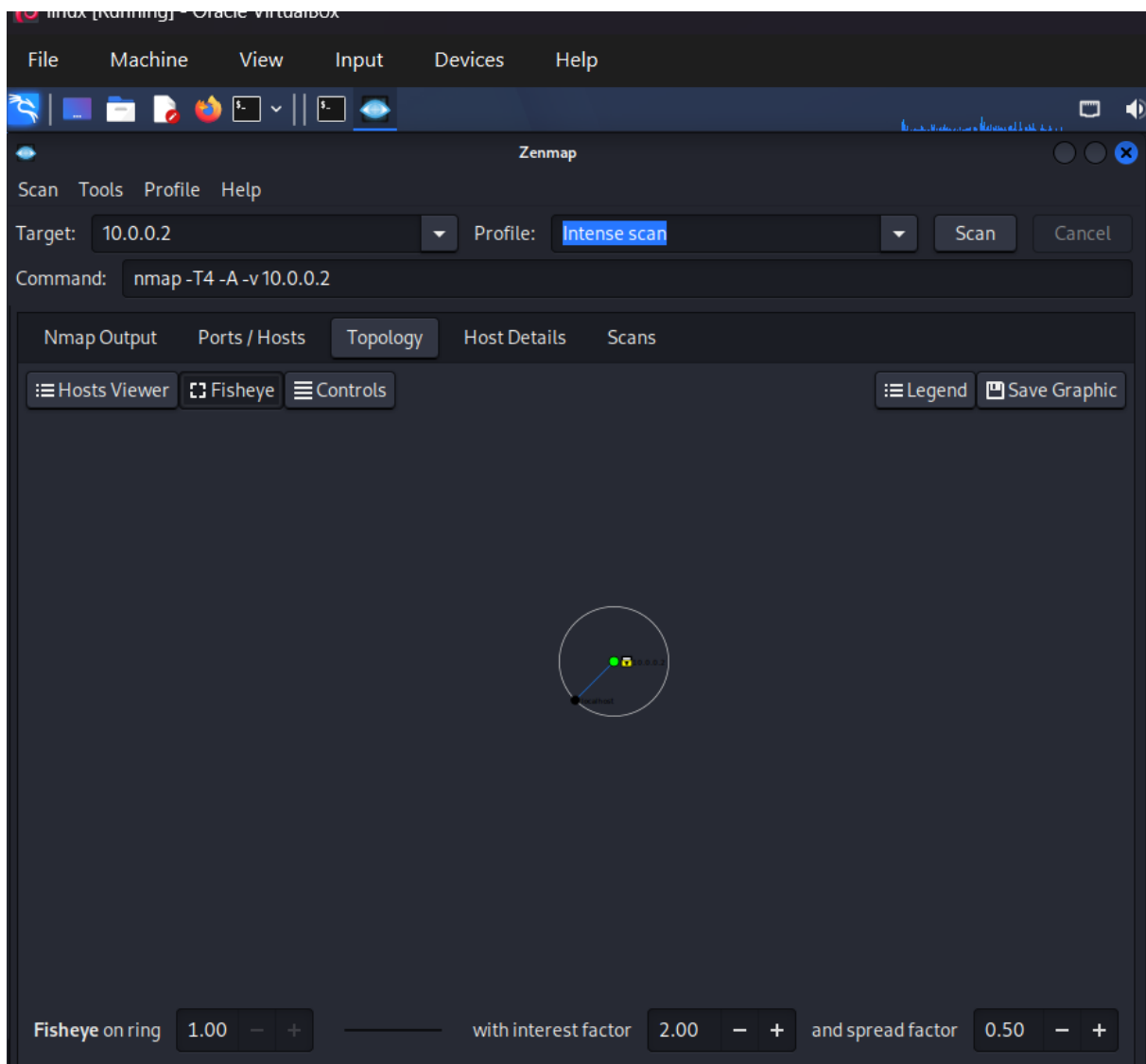
Step 3: Host Discovery

Command executed:

```
ping -c 5 10.0.2.15
```

Result: Host responded to ICMP echo requests.

Observation: The host is alive and reachable.



Zenmap topology view confirming the host was discovered on the network.

Step 4: Port Scanning (Hosts and Ports)

Command executed:

```
nmap -A 10.0.2.15
```

Result: No open ports detected.

Possible reasons: firewall enabled or services inactive.

Observation: Host is reachable but no active TCP/UDP services are visible externally. This indicates a hardened system or restricted network environment.

```

(ismail@kali)-[~]
$ nmap -sV 10.0.2.2
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 22:32 +0530
Nmap scan report for 10.0.2.2
Host is up (0.0024s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE      VERSION
80/tcp    open  http         Microsoft IIS httpd 10.0
135/tcp   open  msrpc        Microsoft Windows RPC
445/tcp   open  microsoft-ds?
5357/tcp  open  http         Microsoft HTTPAPI httpd 2.0 (SSDP/UPnP)
27000/tcp open  flexlm       FlexLM license manager
MAC Address: 52:55:0A:00:02:02 (Unknown)
Service Info: OS: Windows; CPE: cpe:/o:microsoft:windows

Service detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 16.44 seconds

```

nmap -sV service/version scan showing open ports and running services.

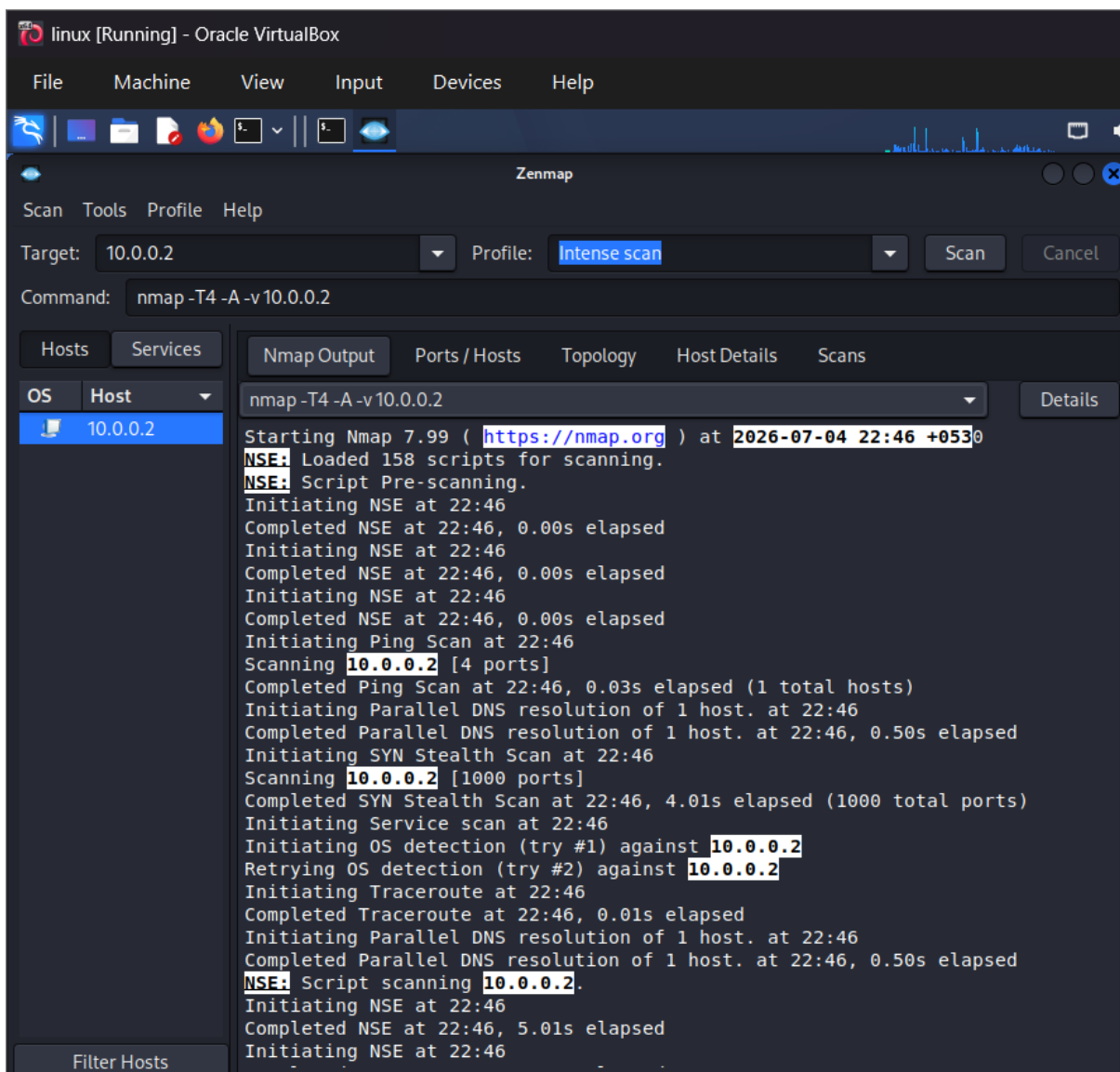
```

(ismail@kali)-[~]
$ nmap -O 10.0.2.2
Starting Nmap 7.98 ( https://nmap.org ) at 2026-07-04 22:38 +0530
Nmap scan report for 10.0.2.2
Host is up (0.00099s latency).
Not shown: 995 filtered tcp ports (no-response)
PORT      STATE SERVICE
80/tcp    open  http
135/tcp   open  msrpc
445/tcp   open  microsoft-ds
5357/tcp  open  wsddapi
27000/tcp open  flexlm0
MAC Address: 52:55:0A:00:02:02 (Unknown)
Warning: OSScan results may be unreliable because we could not find at least 1 open and 1 closed port
Aggressive OS guesses: AT&T BGW210 voice gateway (94%), Oracle Virtualbox Slirp NAT bridge (90%), QEMU user mode ne
twork gateway (89%), HP 9100c Digital Sender printer (J3113A) (86%), Minolta Di550 laser printer (85%), NEC SuperSc
ript printer (85%), Cisco SG 500 switch (85%), Yamaha RX-V2067 or RX-V3900 audio receiver (85%)
No exact OS matches for host (test conditions non-ideal).
Network Distance: 1 hop

OS detection performed. Please report any incorrect results at https://nmap.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 8.33 seconds

```

nmap -O OS detection scan with aggressive OS guesses for the target.



Zenmap Intense scan (nmap -T4 -A) output, including NSE script scanning.

Step 5: Wireshark Installation

Wireshark installed successfully using the package manager.

Run command:

```
sudo wireshark
```

GUI warning ignored (registration issue).

Step 6: Wireshark Traffic Analysis

Wireshark started with sudo wireshark.

Interface eth0 selected.

Traffic generated using:

```
ping -c 5 google.com
```

Wireshark capture started; ICMP packets successfully detected.

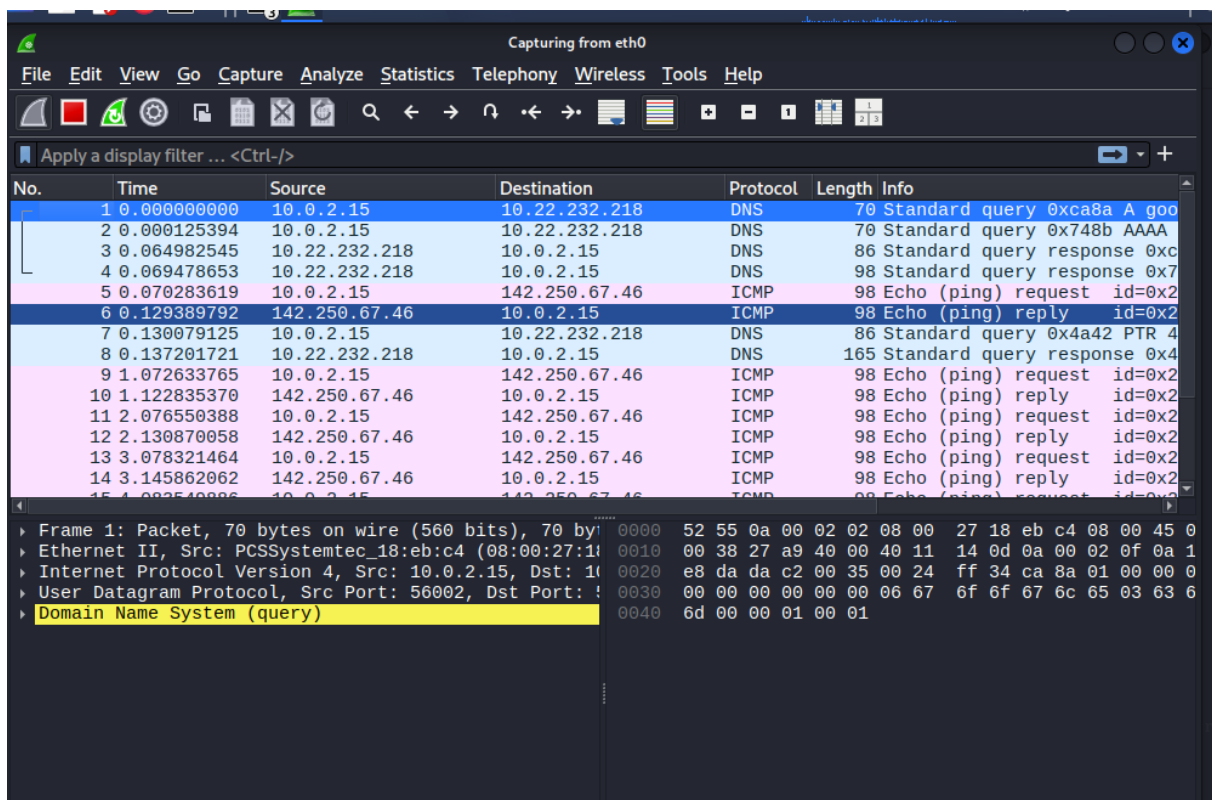
Filter applied:

```
icmp
```

Capture stopped and screenshots saved (overall packets + filtered view).

Observation:

- Host is reachable (ICMP allowed).
- Network communication visible.
- Wireshark capture successful and traffic analysis complete.



Wireshark capture on eth0 showing DNS queries and ICMP echo request/reply traffic.

Step 7: Vulnerability Check

Findings:

- Zenmap Scan: No open ports detected — firewall enabled or services inactive.
- Wireshark Analysis: ICMP packets captured — host reachable, ICMP echo replies allowed.

Vulnerability Analysis:

- No open ports means the external attack surface is reduced.
- However, ICMP being allowed means attackers can perform ping sweeps to discover hosts.

- If insecure protocols (FTP, Telnet, HTTP without SSL) were detected, they would be considered vulnerabilities.

Recommendations:

- Disable ICMP echo replies or use IDS/IPS to block reconnaissance.
- Tighten firewall rules.
- Use secure protocols (HTTPS, SSH).
- Perform regular vulnerability scans.

Step 8: Conclusion

The lab results show that the target host is reachable and ICMP traffic is allowed. Zenmap scan revealed no open ports, which indicates firewall hardening. Overall, the system's external attack surface is limited, but ICMP replies create a reconnaissance risk. Proper firewall configuration and secure protocols are recommended to strengthen security.